Definition

Time series forecasting involves analyzing data that evolves over some period of time and then utilizing statistical models to make predictions about future patterns and trends. It takes into account the **sequential nature of data,** where each observation is dependent on previous observations.

The significance and practical uses of time series analysis

1. **Forecasting/Prediction:** Through the use of existing historical data, time series analysis enables the prediction of future values and trends. The insight obtained from the forecasts helps businesses and organizations make informed decisions and plan their available resources accordingly based on predicted demand.
2. **Trend Analysis:** Time series analysis helps us find and understand patterns in data that change over time. It shows

us how things are changing and how much they are changing. This information is important especially for stakeholders when making decisions and planning for the future.

3. **Seasonal Patterns:** Time series analysis is a helpful tool that allows us to find and understand patterns that repeat over time. For example, in industries like retail, tourism, and agriculture, where things change depending on the season or specific times of the year, time series analysis helps us predict and plan for those changes. It helps businesses know when to expect high or low demand and when to produce or stock up on certain products.
4. **Anomaly Detection:** Time series analysis is a useful tool that can help us find unusual things in data. It can detect when something strange or unexpected happens, like errors or unusual events. This is important because it allows us to catch fraud, keep an eye on how well a system is working, or notice when something is behaving strangely. Time series

analysis helps us find these abnormal behaviors in different situations and applications.

Eg:

1. Banking: Detecting Fraudulent Transactions

🧠 **Concept:**

Banks track customer transactions over time (e.g., amounts, time, frequency). Any sudden change in this pattern can signal **fraud**.

💡 **Example:**

A customer usually spends less than ₹5,000 daily. Suddenly, three transactions of ₹50,000 each occur at midnight from a new country.

🔍 **How Time Series Helps:**

- The system tracks transaction **amount over time**.
- It builds a model of "normal" behavior.
- Sudden spikes or unusual patterns trigger an **anomaly alert**.

🔧 **Models Used:**

- Moving averages
- Seasonal Hybrid Extreme Studentized Deviate (S-H-ESD)
- Isolation Forests with time stamps

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2. IoT (Internet of Things): Spotting Faulty Sensor Data

🧠 **Concept:**

In IoT systems (e.g., smart factories), sensors constantly send data like temperature, pressure, humidity, etc.

💡 **Example:**

A temperature sensor in a machine typically shows 60–65°C. Suddenly, it reports 90°C for 5 minutes.

🔍 **How Time Series Helps:**

- The model knows the **expected range and seasonality** of sensor readings.
- Sudden deviation = potential fault, overheating, or sensor failure.
- Early detection prevents damage.

🔧 **Models Used:**

- ARIMA/Seasonal ARIMA (SARIMA)
- LSTM (Long Short-Term Memory) for sequences
- Prophet for trend + seasonality

5. Financial Analysis: Time series analysis is a valuable tool used in finance to study stock prices, predict market trends, manage investment portfolios, and assess risks. It helps investors and financial institutions make smarter decisions by analyzing past patterns and trends in the stock market. By understanding how prices have changed over time, investors can gain insights into potential future movements and adjust their strategies accordingly. This analysis helps in identifying opportunities for profitable investments and managing risks associated with market fluctuations.

6. Environmental Monitoring: Time series analysis is used in environmental sciences to study and predict things like weather patterns, air quality, water levels, and other environmental factors. It helps scientists understand how these variables change over long periods of time, identify patterns and trends, and make informed decisions to manage and protect the environment. For example, by analyzing historical data, researchers can predict future climate patterns, assess the impact of pollution on air quality, or monitor water levels in rivers and lakes. This information is crucial for environmental management and decision-making to ensure the well-being of our planet.

This is not an exhaustive list of the applications of Time Series forecasting. There are many other scenarios which leverage Time Series Forecasting.

Understanding Time Series Data

Characteristics of time series data

Time series data consists of recorded observations that are associated with specific timestamps, allowing us to understand how variables change over time.

Some key characteristics of time series data include

1. Temporal Ordering (Happens in Order by Time)

👉 **Simple Explanation:**

Time series data is recorded in the **order of time** — what happened first, then next, and so on.

📊 **Example:**

Let’s say you're tracking your daily steps:

Day	Steps
Monday	5000
Tuesday	5500
Wednesday	5200

You must keep the order — you can't shuffle the days. The order matters because we want to see how your steps change **over time**.

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2. Time Dependency (Past Affects the Present)

👉 **Simple Explanation:**

What happened **before** affects what happens **now**. The values are connected.

📊 **Example:**

You feel tired today because you only walked 2,000 steps yesterday instead of your usual 5,000. That tiredness might make you walk even less tomorrow.

So, today's value (steps, sales, temperature, etc.) is related to yesterday's value. That's **time dependency**.

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3. Irregular Sampling (Uneven Time Gaps)

👉 **Simple Explanation:**

Sometimes data is not collected at regular time intervals — and that makes analysis harder.

📊 **Example:**

Suppose your smart watch records heart rate like this:

Time	Heart Rate
10:00 AM	72
10:02 AM	74

10:08 AM	120 (running)
10:30 AM	78

Notice how the time gaps are not even—2 mins, then 6 mins, then 22 mins. That's called **irregular sampling**.

To analyze it properly, you may need to **fill in missing data** or adjust the method to handle the uneven timing.

Components of time series: trend, seasonality, and noise

We can break down time series data into three primary components, which aid in comprehending the underlying patterns:

1. Trend = Overall direction over time

👉 **Simple Meaning:**

A **trend** shows if the data is generally going **up**, **down**, or staying the same **over a long time**.

Trends can be linear (constant increase or decrease) or nonlinear (curved or oscillating) or staying the same.

📊 **Example:**

Imagine you're tracking your height each year as a child:

Year	Height (cm)
2015	120
2016	125
2017	130

2018	135

You can see a clear upward **trend** — you're growing taller every year.

2. Seasonality = Repeating patterns

👉 **Simple Meaning:**

Seasonality is when something **repeats in a pattern** at regular time intervals — like every day, month, or year.

📊 **Example:**

You sell ice cream:

Month	Sales
Jan	200
Feb	250
Jun	1000
Jul	1100
Dec	300

Every **summer**, sales go **way up**, and in **winter**, they go down. This repeating pattern is called **seasonality**.

3. Noise = Random changes

👉 **Simple Meaning:**

Noise is all the small **random changes** in the data that you **can't predict** or **don't have a reason** for.

📊 **Example:**

Imagine your daily steps:

Day	Steps
Monday	5000
Tuesday	5200
Wednesday	3000 ← maybe you were sick (noise)
Thursday	5100

That sudden drop on Wednesday is **noise** — a random change that doesn't follow the usual pattern. It could be due to anything — weather, mood, or being sick.

🧠 Summary in 1 Line Each:

- **Trend**: Long-term direction (e.g., growing taller over years).
- **Seasonality**: Repeating patterns (e.g., ice cream sells more every summer).
- **Noise**: Random changes (e.g., walked less one day for no clear reason).

Stationarity and its significance in time series analysis:

Stationarity is a fundamental concept in time series analysis.

Stationarity refers to the condition where the statistical properties of a time series, such as its mean, variance, and autocorrelation, remain consistent over time.

✅ Example of Stationary Data:

Let's say you measure your room temperature every hour and it's always around 25°C with small ups and downs:

Hour	Temp (°C)
1	25
2	24.8
3	25.2
4	25.1
5	24.9

- The average stays close to 25°C.
- The changes are small and random.
- This is **stationary**.

❌ Example of Non-Stationary Data (with Trend):

Now imagine you're tracking your savings over months:

Month	Savings (₹)
Jan	1000
Feb	1200
Mar	1400
Apr	1600

- The amount keeps increasing — that's a **trend**.
- The **mean changes over time**.
- This is **non-stationary**.

❌ Example of Non-Stationary Data (with Seasonality):

Let's say you record how many people visit your shop every month:

Month	Visitors

Jan	100
Feb	110
Jun	300
Jul	320
Dec	90

- Visitor count **repeats in a seasonal pattern** (summer ↑, winter ↓).
- The **mean and variance** change based on the season.
- This is also **non-stationary**.

🛠️ How to Make Data Stationary?

- **Remove trend**: Use **differencing** (subtracting previous values).
- **Remove seasonality**: Use **seasonal differencing** or **decomposition**.
- **Transformations**: Use log or square root to stabilize variance.

Stationarity is significant because of the following:

1. **Simplified Analysis:** Stationary time series display consistent statistical properties, simplifying their analysis and modeling. Techniques and models designed for stationary data are known for their reliability and accuracy.
2. **Reliable Forecasts:** Stationary time series data typically displays consistent patterns, which simplifies the process of forecasting future values. Models developed using stationary data are known to offer more dependable and precise predictions.
3. **Statistical Assumptions:** Assumptions of stationarity are made by several time series models, including ARMA and ARIMA. Deviating from this assumption can result in unreliable outcomes and inaccurate predictions.
4. **Trend and Seasonality Analysis:** By achieving stationarity in the data, we can effectively distinguish the trend and seasonality components, enabling us to analyze and model these patterns independently.

🔄 What does it mean to "remove trend and seasonality"?

Let’s imagine your time series data is like a cake 🎂 made up of 3 layers:

1. **Trend layer** – the overall growth or decline (like your savings going up over time).

2. **Seasonality layer** – repeating patterns (like more sales every December).

3. **Noise layer** – random ups and downs (like sudden one-day drop in sales).

When we say **“remove trend and seasonality”**, we mean:

> Strip away the first two layers — the trend and the seasonality — so only the **"noise"** is left.

This remaining data is called **stationary data** — it has no clear upward/downward trend or repeating pattern.

📉 Example

Imagine this sales data for a small shop:

Month	**Sales**
Jan	100
Feb	120
Mar	140
Apr	160

May	300 ← seasonal spike
Jun	180
Jul	200

You can see:

- Sales are **slowly increasing over months** → that's the **trend**.
- A big jump in **May** every year → that's **seasonality** (maybe a festival).

✂️ Step 1: Remove the Trend

You subtract each value from the previous one. This is called **differencing**.

New series (difference from last month):

Month	Change
Feb	+20
Mar	+20
Apr	+20
May	+140 ← Big jump!
Jun	-120
Jul	+20

Now the **overall upward trend is gone**, but the seasonal spike in May is still there.

✂️ **Step 2: Remove Seasonality**

You do **seasonal differencing**: subtract this year’s May from last year’s May, and so on.

This helps remove the repeating seasonal pattern — now you're left with **data that has no trend and no repeating pattern**.

🔍 How to Analyze Them One by One

1. Trend Analysis:

Before differencing, you can **fit a line** to your data.

- Is it going up (positive trend)?
- Going down (negative trend)?
- Staying flat?

👉 Helps you know whether your business, temperature, etc., is growing or shrinking over time.

2. Seasonality Analysis:

You can **group by month or season** and see:

- Do sales always go up in May?
- Are winters always slow?

👉 Helps you plan for **seasonal demands** or fluctuations.

In practical terms, making time series data stationary may involve making changes or using techniques to eliminate trends or seasonality. Stationarity is an important factor to consider when working with time series data to ensure accurate analysis and dependable forecasts.

Overview of Time Series Models

Time series models are statistical tools that experts use to study and predict data that changes over time. These models help us uncover patterns, trends, and relationships in the data, which in turn allows us to make informed predictions about what might happen in the future.

Below is a brief overview of some commonly used time series models:

Moving Average (MA) Model: This model calculates the average of past observations with the aim of predicting future values. It is useful for capturing short-term fluctuations and random variations in the data.

- **Assumptions:** The observations are a linear combination of past error terms, and there is no autocorrelation between the error terms.
- **Parameters:** The order of the model (q) determines the number of lagged error terms to include.

- **Strengths:** MA models are effective in capturing short-term dependencies and smoothing out random fluctuations in the data.

🧮 Formula (in simple terms):

MA(q) = mean + (error at time t-1 × weight) + (error at time t-2 × weight) + ... up to q lags

Where:

- **q** = how many past errors you use

- **error** = difference between actual value and predicted value

$MA(q)=\mu+\theta_1 e_{t-1}+\theta_2 e_{t-2}+\cdots+\theta_q e_{t-q}+e_t$

μ - It is the overall average (mean) level of the time series.

Autoregressive (AR) Model: This model predicts future values based on a linear combination of past observations.

- **Assumptions:** It assumes that the future values depend on the previous values, capturing long-term trends and dependencies.
- **Parameters:** The order of the model (p) determines the number of lagged observations to include.
- **Strengths:** AR models are useful for capturing long-term dependencies and trends in the data.

Formula (for AR(p) model):

$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + e_t$

Where:

- **y_t is the value we want to predict (e.g., today's temperature)**

- **$y_{t-1}, y_{t-2}, \ldots\ldots$ are past values (e.g., yesterday's, the day before…)**

- **$\phi_1, \phi_2, \ldots$ are weights (how much past values matter)**

- **c is a constant (baseline)**

- **e_t is a random error (surprise or noise)**

Autoregressive Moving Average (ARMA) Model

The ARMA model is a combination of two models:

- **AR (Autoregressive): Uses past values to predict future ones.**

- **MA (Moving Average): Uses past errors (unexpected variations) to improve predictions.**

Together, ARMA captures both long-term trends and short-term fluctuations in a stationary time series.

🔧 **Assumptions:**

- **The current value is a linear combination of:**
  - **past observations (AR part)**
  - **past error terms (MA part)**
- **The error terms are uncorrelated (no autocorrelation).**

$Cov(e_t, e_{t-k})=0$ for any lag $k=0$ ie no trend or seasonality followed

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⚙️ **Parameters:**

- **p: Order of the AR component (number of past observations used)**

- **q: Order of the MA component (number of past error terms used)**

ARMA (Autoregressive Moving Average) Model: Explained with Equation and Example

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- ✅ **Final General Equation for ARMA(p, q):**

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=1}^{q} \theta_j e_{t-j} + e_t$$

Where:

- y_t: Value at time t
- c: Constant
- ϕ_i: AR (autoregressive) coefficients, i.e., how much past values influence the current value
- θ_j: MA (moving average) coefficients, i.e., how much past errors influence the current value
- e_t: Error term (white noise) at time t

🧲 Example: ARMA(2, 1)

Suppose:

- **p=2: Use the last two observations**

- **q= 1: Use one previous error**

This means:

- **You're using the last two values of the time series to predict the current value (AR part).**
- **You're also using the last error to improve the prediction (MA part).**
- **A new error is always present to account for randomness.**
- **The constant c acts as a baseline level for predictions.**

🧠 What is ARIMA?

ARIMA stands for: **Autoregressive Integrated Moving Average**

- **AR** = Autoregressive (uses past values)
- **I** = Integrated (means differencing to remove trends)
- **MA** = Moving Average (uses past errors)

So, **ARIMA** is a model that predicts future values by combining

1. **Past values** of the data (AR part),
2. **The trend** removed by **differencing** (I part),
3. **Past mistakes/errors** in prediction (MA part).

🔄 **Why use ARIMA?**

Because real-world data often has:

- **Trends** (e.g., sales go up every year)

- **Seasonality** (e.g., higher sales every December)

- **Fluctuations** (random ups and downs)

But ARMA models only work well if the data is **stationary** (no trend or changing pattern).

So ARIMA **first makes the data stationary** by subtracting the previous value (called **differencing)** and then uses ARMA on the cleaned-up version.

📊 Imagine this data:

Time (t)	Value (y_t)
1	10
2	12
3	14
4	16
5	18

You can see this data has a clear **upward trend** — it increases by 2 each time.

🔧 Step: Remove trend using **first-order differencing**

Differencing means subtracting the previous value from the current value:

$$\text{Differenced value at time } t = y_t - y_{t-1}$$

Let's apply this:

Time (t)	y_t	Differenced ($\Delta y_t = y_t - y_{t-1}$)
2	12	12 - 10 = **2**
3	14	14 - 12 = **2**
4	16	16 - 14 = **2**
5	18	18 - 16 = **2**

❄️ What is SARIMA?

SARIMA = **Seasonal ARIMA**

It’s just like **ARIMA**, but it also handles **seasonal patterns** — things that repeat at regular time intervals like:

- Monthly sales peaking every December

- Daily electricity usage dropping at night

- Weekly traffic being higher on Mondays

So SARIMA = ARIMA + Seasonal stuff

🧠 **Breakdown:**

SARIMA combines:

- **AR (Autoregressive):** Uses past values.

- **I (Integrated):** Removes trends using differencing.
- **MA (Moving Average):** Uses past errors.

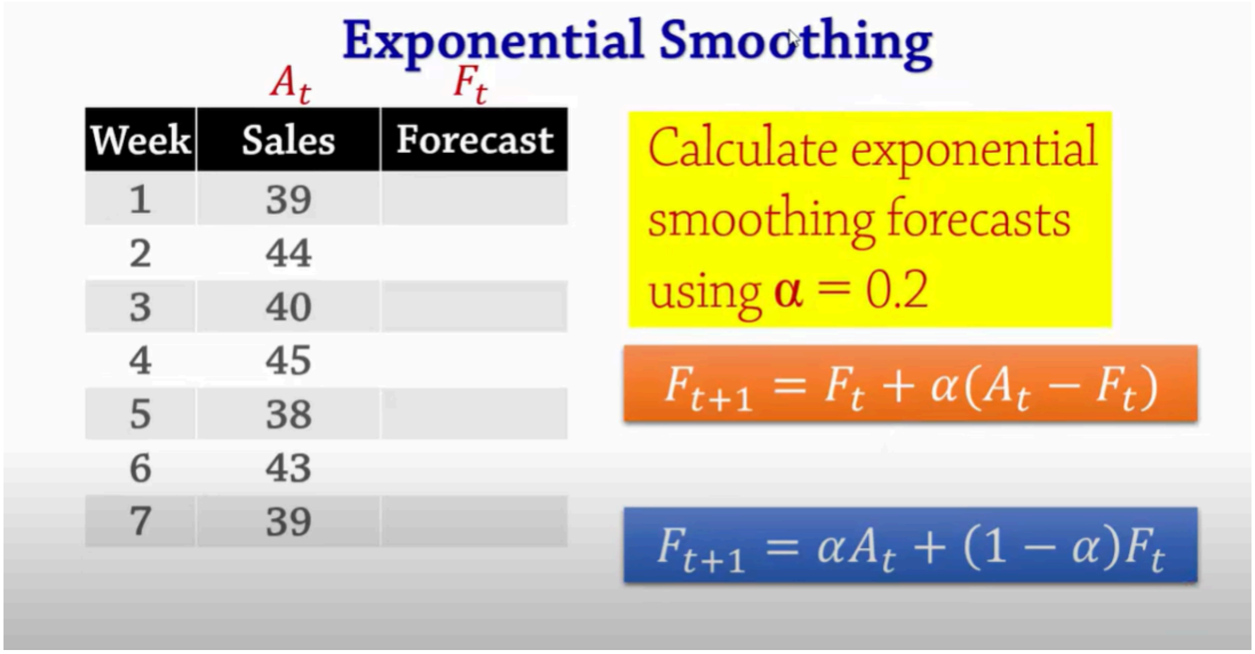
PLUS

- **Seasonal AR (P):** Uses past seasonal values (e.g., value from 12 months ago)
- **Seasonal Differencing (D):** Removes repeating patterns (e.g., subtract current Jan from last Jan)
- **Seasonal MA (Q):** Uses past seasonal errors

⚙️ **Parameters in SARIMA(p, d, q)(P, D, Q, s):**

- **p, d, q** → like ARIMA (non-seasonal part)
- **P, D, Q** → same as above but for **seasonal part**
- **s** → the number of steps in a full season

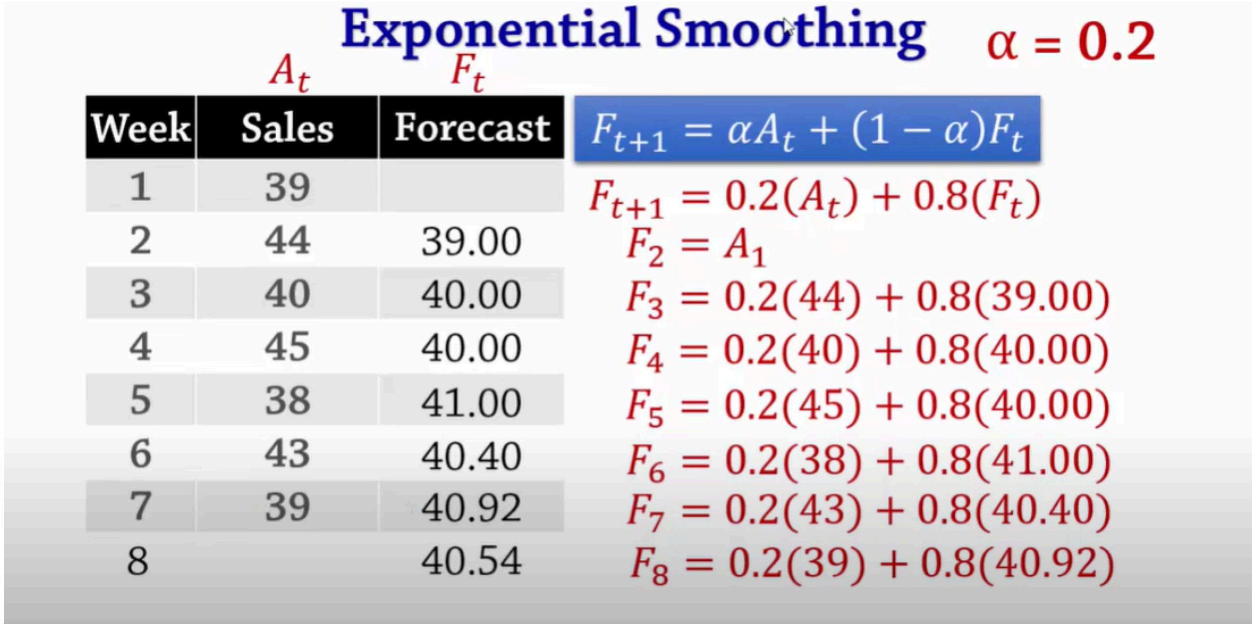
  (e.g., s = 12 for monthly data with yearly seasonality)



Alpha (α): The **smoothing factor** between 0 and 1.

- Closer to **1** → more weight to **recent data**.
- Closer to **0** → smoother curve, slower reaction to new changes.

Exponential Smoothing is a method used to **predict future values** by giving **more importance to recent data** than older data. It's like taking an average — but not a simple one. The most recent data points matter **more**, and the importance **decreases** as we go back in time.



Exponential Smoothing $\alpha = 0.2$

Week	Sales (A_t)	Forecast (F_t)
1	39	
2	44	39.00
3	40	40.00
4	45	40.00
5	38	41.00
6	43	40.40
7	39	40.92
8		40.54

$$F_{t+1} = \alpha A_t + (1-\alpha)F_t$$

$$F_{t+1} = 0.2(A_t) + 0.8(F_t)$$

$$F_2 = A_1$$

$$F_3 = 0.2(44) + 0.8(39.00)$$

$$F_4 = 0.2(40) + 0.8(40.00)$$

$$F_5 = 0.2(45) + 0.8(40.00)$$

$$F_6 = 0.2(38) + 0.8(41.00)$$

$$F_7 = 0.2(43) + 0.8(40.40)$$

$$F_8 = 0.2(39) + 0.8(40.92)$$

Vector Autoregression (VAR) Model: This model is used when multiple time series variables interact with each other, and it captures the relationships and dependencies between variables, making it suitable for macroeconomic forecasting.

🧮 Mini Example Table:

Time	GDP (y_1)	Interest (y_2)
t-2	90	5%
t-1	95	6%
t	?	?

Let's say we have:

- y_{1t} = GDP at time t
- y_{2t} = Interest rate at time t

A VAR(1) model (using 1 lag) would look like:

Equation 1:

$GDP_t = a_1 + b_{11} \cdot GDP_{t-1} + b_{12} \cdot Interest_{t-1} + e_{1t}$

Equation 2:

$Interest_t = a_2 + b_{21} \cdot GDP_{t-1} + b_{22} \cdot Interest_{t-1} + e_{2t}$

⚙️ Parameters:

- **p**: the number of past time steps to use (called the "lag")
- **k**: number of variables in the system

So a **VAR(p)** model with **k variables** includes **p × k** lag terms in each equation.

Machine Learning Models: Machine learning algorithms, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, can also be applied to time series analysis. These models can capture complex patterns and dependencies in the data.

Machine Learning (ML) models — especially **deep learning models** like **RNNs** and **LSTMs** — can:

- Handle **nonlinear** relationships.
- Automatically learn **complex patterns**.

- Work with **non-stationary** data without manual differencing.

Metrics for evaluating time series models

Evaluating the performance of time series models requires the use of specific metrics tailored to the characteristics of time-dependent data. Some commonly used metrics include:

1. **Mean Absolute Error (MAE)**: This metric measures the average absolute difference between the predicted and actual values. It provides a straightforward measure of the model's accuracy.
2. **Root Mean Squared Error (RMSE)**: RMSE calculates the square root of the average squared difference between the predicted and actual values. It penalizes larger errors more heavily than MAE.

3. **Mean Absolute Percentage Error (MAPE):** MAPE calculates the average percentage difference between the predicted and actual values. It provides a relative measure of the model's accuracy.
4. **Forecast Bias:** Forecast bias measures the tendency of the model to consistently overestimate or underestimate the actual values. A bias close to zero indicates a well-calibrated model.

What is Forecast Bias?

- When you use a model to predict future values, **forecast bias** tells you if your model is generally **too high** or **too low** compared to what really happens.
- If the model always guesses numbers **higher** than the actual values, it has a **positive bias** (overestimates).
- If it always guesses numbers **lower** than the actual values, it has a **negative bias** (underestimates).
- If the average difference between predicted and actual values is close to **zero**, it means the model doesn't have a consistent bias and is considered well-calibrated.

Simple Example:

Imagine you predict the daily temperature for a week, and the actual temperatures are:

Day	Actual Temp	Predicted Temp	Difference (Predicted - Actual)
1	20°C	22°C	+2
2	18°C	19°C	+1
3	21°C	23°C	+2
4	19°C	20°C	+1
5	20°C	21°C	+1

- Here, differences are all positive, meaning your model tends to **overestimate** temperatures.

- The average difference (bias) = (2 + 1 + 2 + 1 + 1) / 5 = 1.4°C
- This means your model has a **positive forecast bias of +1.4**, consistently predicting temperatures about 1.4 degrees too high.

Why does bias matter?

- If the bias is large (positive or negative), your model's predictions are **systematically off** in one direction, which can be problematic for decision making.
- A **bias close to zero** means your model is neither consistently over nor underestimating — it's balanced and more reliable.